

# Gradient-Guided Network With Fourier Enhancement for Glioma Segmentation in Multimodal 3D MRI

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**Abstract**—Glioma segmentation is a crucial task in computer-aided diagnosis, requiring precise discrimination between lesions and normal tissue at the pixel level. Popular methods neglect crucial edge information, leading to inaccurate contour delineation. Moreover, global information has been proven beneficial for segmentation. The feature representations extracted by convolution neural networks often struggle with local-related information owing to the limited receptive fields. To address these issues, we propose a novel edge-aware segmentation network that incorporates a dual-path gradient-guided training strategy with Fourier edge-enhancement for precise glioma segmentation, a.k.a. GFNet. First, we introduce a Dual-path Gradient-guided Training strategy (DGT) based on a Siamese network guiding the optimizing direction of one path by the gradient from the other path. DGT pays attention to the indistinguishable pixels with large weight-updating gradient, such as the pixels near the boundary, to guide the network training, addressing hard samples. Second, to further perceive the edge information, we derive a Fourier Edge-enhancement Module (FEM) to augment feature edges with high-frequency representations from the spectral domain, providing global information and edge details. Extensive experiments on public glioma segmentation datasets, BraTS2020 and Medical Segmentation Decathlon (MSD) glioma and prostate segmentation, demonstrate that GFNet achieves competitive performance compared to other state-of-the-art methods, both qualitatively and quantitatively.

**Index Terms**—Glioma segmentation, gradient guidance, fourier transformer, edge information.

## I. INTRODUCTION

GLIOMAS are the most common malignancies in the central nervous system and have a high mortality rate. This disease can be classified into two categories according to the World Health Organization (WHO) grading system [1]: low-grade gliomas (LGGs) and high-grade gliomas (HGGs) [2]. Gliomas typically contain heterogeneous sub-regions, including edema, enhancing core, and non-enhancing core. As the shape and location of gliomas can be highly diverse, the glioma segmentation by clinicians requires a substantial amount of time and expertise, making it susceptible to subjective factors. Automatic segmentation algorithms can enhance diagnostic accuracy and speed, aiding in the early detection and treatment of gliomas. Additionally, they provide clinicians with precise glioma segmentation results, serving as decision support tools and facilitating the development of improved treatment plans. Therefore, accurate automatic glioma segmentation is a very active area of research interest.

Deep Convolutional Neural Networks (DCNNs) have demonstrated significant efficacy in the field of image segmentation, attributed to their superior feature extraction capabilities. They have been pivotal in extracting complex features from high-dimensional data, subsequently improving the precision of image classification and segmentation tasks. Two prevalent network architectures, renowned for their effectiveness in addressing the image segmentation problem: Fully Convolutional Networks (FCNs) [3] and U-Net [4]. FCN-based algorithms [5], [6], [7] can adapt to images of arbitrary resolution. U-Net is equipped with an Encoder-Decoder architecture and skip-connection operation to combine multi-scale feature maps, resulting in significant performance improvements. Following this “U-shape” paradigm, numerous segmentation methods [8], [9], [10], [11], [12], [13], [14], [15], [16], [17] have been developed for specific medical tasks. However, different from other organs, gliomas exhibit a high degree of diversity in terms of their shape, size, and location, rendering the segmentation task more intricate. Furthermore, gliomas typically feature indistinct or irregular boundaries that intermingle with adjacent normal tissue, while some other types of tumors, such as lung tumors or breast tumors, exhibit relatively clear boundaries in images.

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Achieving precise glioma segmentation is crucial for the reliability of medical image analysis. While existing U-Net-based methods have yielded acceptable results, they face two key limitations: a) Most U-Net-based methods [4], [8], [9], [10], [11], [12], [13], [14], [15] use modified CNN/Transformer architectures as encoder-decoders for medical image segmentation, relying on full supervision between segmentation predictions and ground-truth. However, these methods overlook crucial edge information in regions of interest, especially in areas with weak boundaries such as gliomas and skin lesions, leading to suboptimal segmentation results [18]. Some edge-based approaches attempt to mitigate this problem by using edge information as additional supervised signals [18], [19] and learning relevant edge attention information [20], [21], [22]. Existing edge-based methods use edge loss functions to forcibly constrain networks to perceive edge information, neglecting the inherent correlation between edges and segmentation masks. b) CNN-based methods [4], [8], [9], [10], [11], [12], [13], [14] struggle with local feature maps due to limited receptive fields, failing to perceive global information. With the emergence of Vision Transformer, some Transformer-based algorithms [15], [23], [24], [25], [26] aim to learn long-range dependencies to acquire global representations. However, Transformer-based methods face challenges such as requiring massive computational resources, easily overfitting on limited-scale medical image segmentation datasets, and neglecting local-related information. The gradient information highlights the parts of the input image that are difficult to segment, including low-contrast boundaries.

To address the above problems, we propose GFNet, a novel dual-path gradient-guided network aided by Fourier edge-enhancement for precise glioma segmentation. The use of gradient information to update the weights of convolutional networks via backpropagation is a well-known technique. Ramprasaath et al. [27] introduce a new technique that incorporates gradient information to visualize the specific localization map in feature maps. The gradient information contains useful edge information for segmentation networks. Inspired by this, we propose a gradient-guided training strategy for a Siamese U-shape networks. In the first branch, we extract gradient information from the feature maps of the Encoder's hidden layers. Gradient information highlights challenging areas of the input image, such as weak boundaries. In the second branch, we combine the gradient information with critical feature maps to guide the network's training, directing the network's attention to difficult-to-segment regions. This dual-path gradient-guided training strategy enables to preserve more edge details to improve glioma segmentation performance.

In addition, due to limited receptive fields, convolutional filters can only generate local feature representations and often fail to capture global information. According to the spectral convolution theorem in Fourier theory [28], [29], a feature map in the spatial domain, when transformed by FFT, is globally affected by pixel-wise updates in the spectral domain. To leverage this property, we design the Fourier Edge-enhancement Module (FEM), which maintains global information and reinforces edge information of the feature maps, making the anatomy more discriminative. First, we use FFT to obtain amplitude and phase spectrum in the frequency space. Then, we apply a central-crop

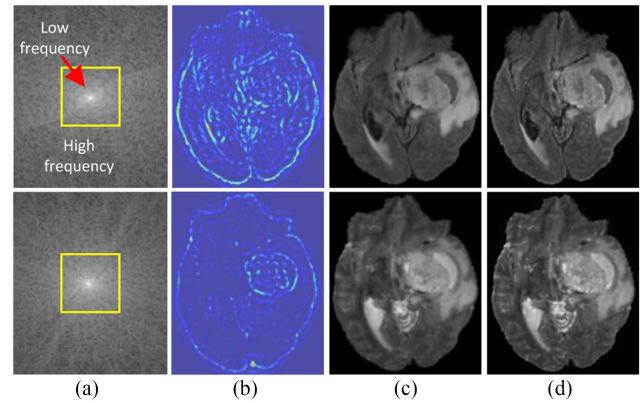


Fig. 1. Illustration of Fourier edge-enhancement. (a) Amplitude spectrum; (b) reconstructed images generated by the phase information and the high-frequency part of amplitude information; (c) original image; (d) augmented image.

window to split the high-frequency and low-frequency parts of the amplitude spectrum. Finally, we combine the high-frequency amplitude spectrum and the original phase spectrum to obtain discriminative feature maps using inverse FFT. The proposed FEM integrates local and non-local information, which is beneficial for precise segmentation. The process is illustrated in Fig. 1.

Our main contributions are summarized as follows:

- We propose a novel Dual-path Gradient-guided Training strategy (DGT) that incorporates the feature map with gradient to enhance the edge information in the training process.
- Fourier Edge-enhancement Module (FEM) is introduced to reinforce the global and detail information of the original features in the spectral domain by integrating the features of high-frequency parts.
- We conduct extensive experiments on three typical multimodal segmentation datasets, i.e., Brain Tumor Segmentation 2020 (BraTS2020), Medical Segmentation Decathlon (MSD) glioma and prostate segmentation, and the proposed GFNet achieves superior performance than other methods, which demonstrates its efficiency.

## II. RELATED WORKS

### A. Medical Image Segmentation

The use of U-shaped networks has yielded promising results in medical image segmentation, leading to the development of various variants such as V-Net [8], Attention U-Net [13], ResU-Net [10], DenseU-Net [30], KiUnet [14], U-Net++ [12], BET-Net [31], VolumeFormer [32], DeSeg [33], and F<sup>2</sup>Net [34] to improve their performance. Attention U-Net [13] introduced the attention gate (AG) model to focus on salient features that benefit to segmentation task. Inspired by [35], Xiao et al. [10] proposed a U-Net-like model combined with the weighted attention mechanism and skip connection. KiUnet [14] adapted an auxiliary overcomplete convolutional architecture that projects the image into higher dimensions to capture detail and edge information. Zhou et al. [12] put forward multi-depth U-Nets aided by a new skip connection scheme, which enhanced the ability for feature extraction. With the success of Vision Transformer in the

field of computer vision, the idea of self-attention was naturally introduced to solve the problem of long-range dependencies in medical image segmentation. There were some representative Vision Transformer-based methods such as TransUNet [15], TransBTS [23], and UNETR [25]. TransUNet [15] proposed a 2D medical image segmentation framework, which deployed several Transformer layers after Encoder to extract the hidden features with global contexts. Wang et al. [23] first explored Transformer in 3D CNN for MRI brain tumor segmentation. UNETR [25] directly took a Transformer encoder to embed 3D volumes and then merged the feature maps from middle layers via skip connection. SwinUNETR [26] introduced a self-supervised pre-training strategy and Swin Transformer-based UNETR for 3D medical image segmentation. nnU-Net [36] is a self-configuring framework for medical image segmentation that adapts its architecture, preprocessing, and training pipeline with the given dataset without manual adjustment. Wang et al. [37] presented a self-distillation method for a Transformer-based UNet, incorporating a multi-scale fusion block to enhance fine-grained detail refinement. Liang et al. [38] developed a Hybrid Deformable Model (HDM), which consists of both inter- and intra-patient deformations. He et al. [39] proposed a statistical deformation model-based data augmentation method that generates diverse and realistic data for volumetric medical image segmentation.

### B. Fourier Transform in CNN

As an analytical transform, Fourier transform (FT) decomposes the functions depending on space or time into functions depending on the spatial frequency or temporal frequency. Oren Rippel et al. [40] introduced spectral pooling that performs dimensionality reduction by projecting onto the frequency basis set and then truncating the representation. SRCLiqueNet [41] reconstructed high-resolution image from its corresponding low-resolution one, which considers that images in the wavelet domain contain textual details. Then, SRL [42] put forward a spectral residual learning framework for achieving fully global receptive field. FFC [29] proposed a novel convolutional operator that has the main hallmarks of non-local receptive fields and cross-scale fusion within the convolutional unit. Inspired by FFC [29], Y-Net [43] combined the features extracted from both spatial and spectral encoders to improve the segmentation performance of OCT images.

### C. Edge-Based Segmentation

There are many methods based on edge information for pursuing precise segmentation. To address the issue of inaccurate segmentation in weak edge regions. Cao et al. [18] proposed a Edge and Neighborhood Guidance Network (EN-Net) that minimizing edge and neighborhood information in the encoder and decoder stages. Attention and Edge Constraint Network (AEC-Net) [19] incorporated attention mechanisms and a edge constraint branch to maintain edge information. Ali Hatamizadeh et al. [21] introduced boundary aware CNNs for medical image segmentation, which designs a edge network

branch and edge-aware loss to extract organ boundary information. Wang et al. [20] designed an edge-attention preservation (EAP) module to focus on edge-related information, which is implemented by gathering multi-layer feature maps of Encoder to generate a boundary prediction. Zhang et al. [22] proposed an Edge-attention guidance Network, namely ET-Net, that learns edge-attention features from Encoder which are utilised to fuse multi-scale features of Decoder. These methods extract features from the intermediate layers and use edge labels for supervision. Unlike existing edge-based segmentation algorithms, GFNet employs gradient and Fourier transformations to extract high-frequency information. It obtains adaptive internal edge information, enhancing the accuracy of edge segmentation. However, most of the existing edge-based segmentation methods directly constrain the edges, enabling the network to perceive edge information. The proposed GFNet can capture edge information by high-frequency part of feature maps and internal gradient information, which promotes the ability to perceive edges.

## III. METHOD

The framework of the proposed GFNet is illustrated in Fig. 2. Our GFNet consists of two main contributions, including a Dual-path Gradient-guided Training strategy (DGT) and a Fourier Edge-enhancement Module (FEM). First, we train the segmentation network in a gradient-guided manner, which is performed on a Siamese U-Shape network. One branch is a vanilla U-shape network that computes the gradient information, and the other branch integrates the generated gradient with feature maps from middle layers to force the network focus on the areas beneficial to the segmentation. Second, FEM augments the feature map with its corresponding high-frequency part to maintain the contextual details and global information.

### A. Basic Network

A Siamese U-shape network is adopted as the backbone network. Henry et al. [44] designed an efficient Encoder-Decoder segmentation pipeline based on a 3D U-Net. There are four down-sampling blocks in the Encoder and each down-sampling block consists of two groups of a 3D convolutional layer with a kernel size of  $3 \times 3 \times 3$ , a batch normalization layer and an activation layer. Max-pooling layers are placed between down-sampling blocks to reduce the spatial size of feature maps. The Decoder, with symmetric architecture as Encoder, upsamples the feature maps to the original size via trilinear interpolation operations. Skip-connection operation is performed to preserve detailed information. The feature maps  $f_i^e$  of the Encoder are concatenated with their corresponding Decoder's feature maps  $f_i^{de}$ .

### B. Dual-Path Gradient-Guided Training Strategy

Backpropagation algorithm is widely used for training neural networks. The gradient of the loss function, which is associated with the weights of the neural network, is computed to update the network's parameters. In this regard, the gradient information can be utilized to further augment the edge information from



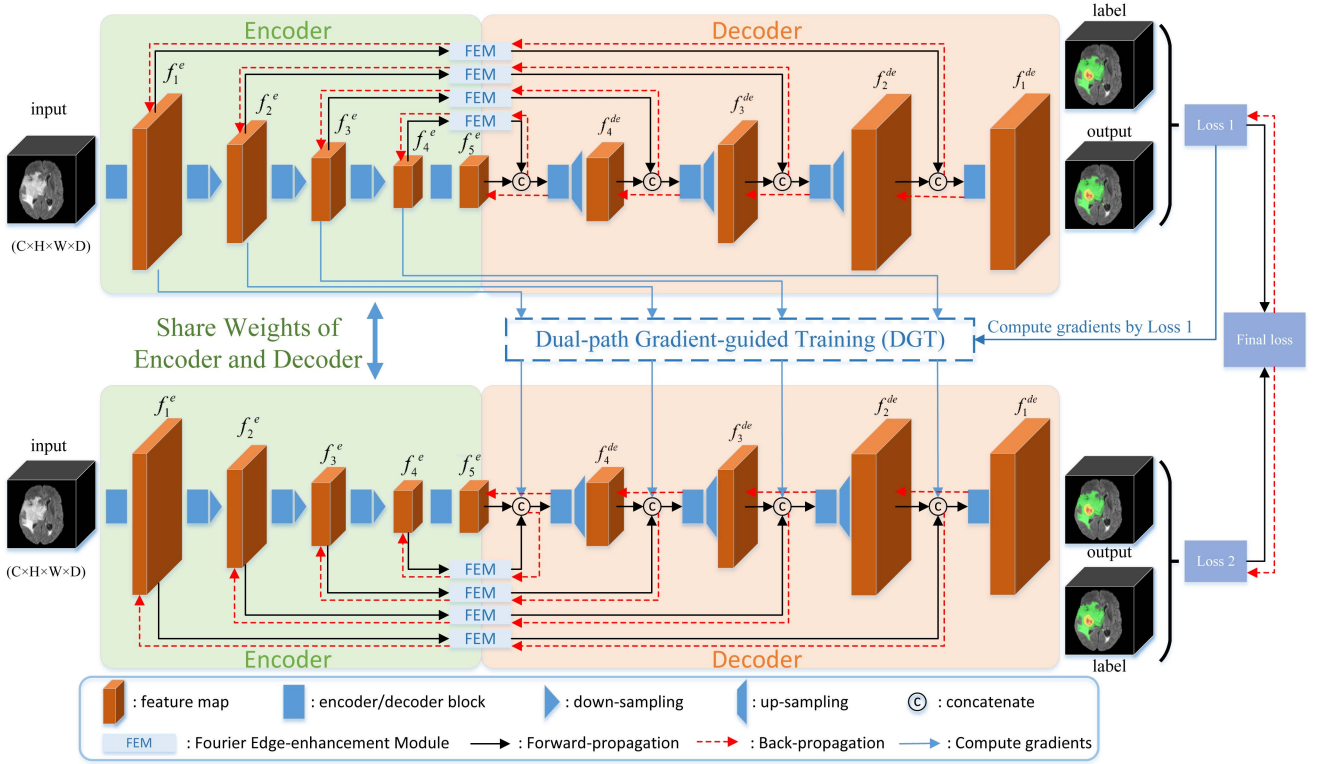


Fig. 2. Overview of the proposed GFNet. Our dual-path gradient-guided network and Fourier edge-enhancement take modified U-Net [44] as the backbone for glioma segmentation. To maintain detailed and global information, FEM is proposed to replace the skip-connection operation. Furthermore, DGT combines the gradient with hidden layer feature maps in the training phase, making the network training focus on the areas with large deviations.

the perspective of network training. The gradient information reflects the parts of the input image that are difficult to segment, including weak boundaries. Therefore, we propose a novel training strategy that is guided by the gradient. In the training phase, given a multimodal 3D volume  $X \in \mathbb{R}^{C \times H \times W \times D}$  ( $C$  represents the number of modalities,  $H$ ,  $W$ ,  $D$  represent height, width, and depth, respectively), We take  $X$  as the input for two parallel branches of U-Net, which share the same network parameters. The loss is computed with the segmentation prediction and its corresponding ground-truth of the first branch denoted as “Loss 1” in Fig. 2. Then the gradients of feature maps  $f_i^e$  can be obtained. After visually analyzing the gradients derived from the output in Fig. 7, we observed that they exhibited relatively small values. As a result, our initial step involved applying channel-wise normalization to these gradients, which is formulated as:

$$g_i(c, h, w, d) = \left| \frac{\partial L_1}{\partial f_i^e(c, h, w, d)} \right|, \quad (1)$$

and

$$\alpha_i^e(c, h, w, d) = \frac{g(c, h, w, d)}{\sqrt{\sum_c (g_i^2(c, h, w, d))}}, \quad (2)$$

where  $\alpha_i^e$  denotes the corresponding gradient weight of the feature map that is crucial to the segmentation task and requires more attention.  $g_i$  represents the gradient of feature map  $f_i^e$  with

the loss value  $L_1$ . To eliminate negative gradients, we take the absolute values, denoted as  $|g|$ . We found that gradient outliers can occur during the initial stages of training the DGT network. These gradients arise from high loss values at the start of training, resulting in large gradient values during backpropagation. As the loss values decrease, the abnormal gradients gradually disappear. To address this, we first normalize the gradients using the  $L_2$  norm across channels, mitigating the impact of abnormal gradients during initial training. Then, the important weights are utilized to supervise the second branch’s training. We combine the neuron important weight with feature maps of Decoder in the second branch using the following formulation:

$$f_i^{grad} = \alpha_i^e + f_i^e. \quad (3)$$

A large neural important weight indicates that the specific channel of feature map is more sensitive to the loss and plays a more important role in prediction. As a result, it makes sense to guide the network’s training to concentrate on the pivotal segmentation parts. It is worth noting that the weights of the network are not updated when computing the gradients in the first branch. The gradients are only utilized to enhance feature extraction in the second branch. After computing the total loss between the predictions and labels of both branches, backpropagation is performed to update the weights of the segmentation network.

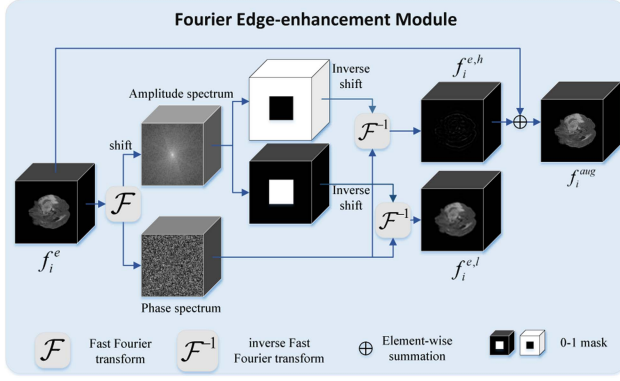


Fig. 3. Framework of Fourier Edge-enhancement Module.

### C. Fourier Edge-Enhancement Module

In this section, we introduce Fourier Edge-enhancement Module (FEM). Two desiderata are considered to design FEM. First, a majority of segmentation methods follow the U-shape architecture that takes convolutional layers as the basic units. Limited receptive field in convolution operation makes the extracted features tend to be local-related. Second, the boundaries of gliomas are normally with weak boundary and irregularly shape, and the original skip-connection operation in U-Net architecture cannot well preserve the edge information. Fast Fourier Transform (FFT) can transfer feature maps in the spatial domain into the spectral domain that contains the global information. High-frequency part of the amplitude spectrum is utilized to enhance the segmentation boundaries. By applying FFT to the feature maps, we convert spatial domain information into the frequency domain. This transformation separates the amplitude and phase spectra, which represent global and local information, respectively. FFT treats the entire feature map as a whole, with each frequency component dependent on all the pixels of the feature map. Thus, the Fourier transform captures the global information of the feature maps. Consequently, the proposed FEM can provide global information and edge details to enhance the segmentation performance. The framework of Fourier Edge-enhancement Module is shown in Fig. 3.

To be specific, given multi-scale feature maps  $f_i^e \in \mathbb{R}^{C \times H \times W \times D}$  from the Encoder, we perform FFT  $\mathcal{F}$  to obtain the frequency space signal of feature maps  $f_i^e$ .  $H, W, D$  represent the width, height and depth of the 3D volume, respectively. The process can be written as:

$$\mathcal{F}(f_i^e)(c, u, v, k) = \sum_{h,w,d}^{H,W,D} f_i^e(h, w, d) e^{-j2\pi(\frac{h}{H}u + \frac{w}{W}v + \frac{d}{D}k)}, \quad (4)$$

where  $j^2 = -1$ . After the transform, an amplitude spectrum  $\mathcal{A}_i^e \in \mathbb{R}^{C \times H \times W \times D}$  and a phase spectrum  $\mathcal{P}_i^e \in \mathbb{R}^{C \times H \times W \times D}$  can be obtained respectively by decomposing a frequency space signal  $\mathcal{F}(f_i^e) \in \mathbb{R}^{C \times H \times W \times D}$ . The result of FFT is typically a complex number, represented as  $\mathcal{F}(f_i^e) = R + jI$ , where  $R$  and

### Algorithm 1: Training Processes of GFNet.

**Training Input:** Volumes  $X$  and labels  $Y$ , Encoder  $E_\theta$ , Decoder  $D_\phi$

**Output:** The trainable parameters of Encoder and Decoder:  $\theta$  and  $\phi$ .

**for**  $ite = 1 \rightarrow iterations$  **do**

$\{x_{ite}, y_{ite}\} \sim \{X, Y\}$

Set  $\alpha^e$  to zero

$f_i^e = E_\theta(x_{ite}), i \in \{1, 2, 3, 4\}$

**{First branch}**

**for**  $i = 1 \rightarrow 4$  **do**

$f_i^{aug} = FEM(f_i^e)$

**end for**

$p_{ite}^1 = D_\phi(f_i^{aug}, \alpha^e), i \in \{1, 2, 3, 4\}$

Compute the loss  $L_1$  by (8) and (9)

Get  $\alpha^e$  by (1) and (2).

$f_i^e = E_\theta(x_{ite}), i \in \{1, 2, 3, 4\}$

**{Second branch}**

**for**  $i = 1 \rightarrow 4$  **do**

$f_i^{aug} = FEM(f_i^e)$

**end for**

$p_{ite}^2 = D_\phi(f_i^{aug}, \alpha^e), i \in \{1, 2, 3, 4\}$

Compute the loss  $L_2$  by (8) and (9)

$\theta \leftarrow \theta - \eta \nabla_\theta (L_1 + L_2)$

$\phi \leftarrow \phi - \eta \nabla_\phi (L_1 + L_2)$

**end for**

$I$  are the real and imaginary parts of the feature map in the frequency domain. The amplitude spectrum  $\mathcal{A}_i^e$  is calculated as  $\sqrt{R^2 + I^2}$ , and the phase spectrum  $\mathcal{P}_i^e$  is determined by  $\arctan(I/R)$ .

The central area of amplitude spectrum indicates the low-frequency signal, while the rest part of amplitude spectrum denotes the high-frequency signal including the edge information of original feature map. In order to reinforce the edge, we define a binary mask  $\mathcal{M}_i$  to separate the amplitude spectrum into high and low-frequency parts. We formulate it as:

$$\mathcal{A}_i^{e,l} = \mathcal{A}_i^e * \mathcal{M}_i,$$

$$\mathcal{A}_i^{e,h} = \mathcal{A}_i^e * (1 - \mathcal{M}_i), \quad (5)$$

where the points in the central part of  $\mathcal{M}_i$  are assigned to 1, and the rest are set to 0. Next, the inverse FFT  $\mathcal{F}^{-1}$  is performed with the high frequency parts  $\mathcal{A}_i^{e,h}$  of the amplitude spectrum  $\mathcal{A}_i^e$ .

$$f_i^{e,h} = \mathcal{F}^{-1}(\mathcal{A}_i^{e,h}, \mathcal{P}_i^e). \quad (6)$$

Finally, the edge augmented features with global information can be obtained by element-wise summation of the original features and the edge features:

$$f_i^{aug} = f_i^{e,h} + f_i^e. \quad (7)$$

This results in the features  $f_i^{aug}$  with both global and detail information. It is interesting to note that the proposed FEM is model-agnostic, i.e., it can serve as a plug-and-play module to solve the problem of local-related features caused by limited receptive field.

## D. Loss Function

We train the proposed network in an end-to-end manner for glioma segmentation [45]. We combine Dice loss [8] with cross-entropy loss to supervise the network training. The Dice loss is formulated as:

$$L_{dice}(P, Y) = 1 - \frac{1}{K} \sum_{k=1}^K \left( \frac{2 |P^k \cap Y^k|}{|P^k|^2 + |Y^k|^2} \right), \quad (8)$$

where  $P$  and  $Y$  represent the segmentation result and its corresponding ground-truth with the size of  $K \times H \times W \times D$ .  $K$  denotes the number of different areas in the gliomas, including the Enhanced Tumor, Tumor Core and Whole Tumor. Cross-entropy loss is formulated as:

$$L_{ce}(P, Y) = - \frac{1}{KHWD} \sum_{k,h,w,d}^{K,H,W,D} Y^k(h, w, d) \cdot \log(P^k(h, w, d)). \quad (9)$$

Since the dual-path gradient-guided training strategy is applied to two parallel branches of U-shape network, we optimize the whole network with the summation of two branches' losses:

$$L_{total} = \underbrace{L_{dice}^1 + L_{ce}^1}_{\text{First branch}} + \underbrace{L_{dice}^2 + L_{ce}^2}_{\text{Second branch}}, \quad (10)$$

where  $L_{dice}^1$  and  $L_{ce}^1$  denote the loss functions of the first branch, while  $L_{dice}^2$  and  $L_{ce}^2$  represent the loss functions of the second branch.

After training the proposed model, we use the first branch to validate the segmentation performance, which ensures that our GFNet incurs no additional computational overload during testing. Additionally, we incorporate FEM in the skip-connections to enhance the edge information perception during inference.

## IV. EXPERIMENTS

In this section, we first introduce the datasets used for validation, evaluation metrics, and implementation details. Then a comparison with several methods is conducted to demonstrate the advancement of our GFNet. Finally, we perform ablation studies to verify the effectiveness of each module and analyze the influence of hyper-parameters.

### A. Datasets and Metrics

1) **Datasets**: Two public medical image segmentation segmentation datasets are applied to validate the efficacy of our method, including BraTS2020, glioma and prostate segmentation task of Medical Segmentation Decathlon (MSD).

**BraTS2020<sup>1</sup>** [45]: **Brain Tumor Segmentation 2020** (BraTS2020) is a 3D MRI multimodal dataset that derived from a challenge organized in conjunction with the MICCAI 2020. This dataset includes 369 cases of patients for training and 128 cases for testing. Each case of patient contains four modalities (T1, T2, T1ce, Flair) that provide different types of biological

information. For example, it is more contrastive for T2 and Flair MRI in tissue water relaxational properties. T1 MRI shows pathological intratumoral take-up of contrast agents. The size of each case is  $240 \times 240 \times 155$ . There are four classes in the segmentation label, following background (Label 0), necrotic and non-enhancing tumor (Label 1), peritumoral edema (Label 2) and GD-enhancing tumor (Label 4). The comparative assessment of BraTS2020 includes the enhancing tumor region (Label 4), the tumor core region (Label 1 and 4) and the whole tumor region (Label 1, 2, 4) three different areas.

**MSD<sup>2</sup>** [46]: **Medical Segmentation Decathlon** (MSD) contains 10 medical image segmentation tasks for different organs or tumors (e.g., segmentation of glioma, heart, prostate, etc.).

**Glioma segmentation**: We select the glioma segmentation sub-task to evaluate the performance of our method. This 3D MRI multimodal dataset is composed of 484 cases for training and 266 cases for validation, which is larger than that of BraTS2020. The other features like measurements and classes of modalities are the same as BraTS2020. While the labels of glioma segmentation of MSD have slight changes, following background (Label 0), necrotic and non-enhancing tumor (Label 2), peritumoral edema (Label 1) and GD-enhancing tumor (Label 3). The comparative assessment includes Enhanced Tumor (Label 3), Tumor Core (Label 2 and 3) and Whole Tumor (Label 1, 2, 3) three different areas.

**Prostate segmentation**: We use the prostate segmentation task to validate our method's generalization to other segmentation tasks. The MSD prostate segmentation dataset, provided by Radboud University, includes 32 MRI cases with two modalities: T2 and ADC. The dataset is randomly divided into 28 cases for training and 7 for validation. Manual segmentation of the entire prostate is performed on transverse T2-weighted scans (resolution:  $0.6 \times 0.6 \times 4$  mm) and ADC maps (resolution:  $2 \times 2 \times 4$  mm). The two segmented classes are the Peripheral Zone (Label 1) and the Transition Zone (Label 2).

2) **Metrics**: For medical image segmentation, the standard metrics comprise *Dice Similarity Coefficient*, *95% Hausdorff Distance* and *Sensitivity* to quantitatively evaluate the segmentation performance from various aspects.

*Dice Similarity Coefficient* (DSC) measures the overlap between two binary segmentation results, ranging from 0 (no overlap) to 1 (perfect overlap).

$$DSC = \frac{2|P \cap Y|}{|P| + |Y|} = \frac{2TP}{FP + 2TP + FN}, \quad (11)$$

where  $TP$ ,  $FP$ ,  $TN$ , and  $FN$  represent the number of True positive samples, False positive samples, True negative samples, and False negative samples, respectively.  $P$  and  $Y$  denote the predicted results and their corresponding labels.

*95% Hausdorff Distance* (HD95) is the maximum distance of the prediction point set  $P$  to the nearest point in the ground-truth point set  $Y$ . In general, we calculate the 95th percentile of the distances between boundary points in the two sets.

$$HD(P, Y) = \max(h(P, Y), h(Y, P)), \quad (12)$$

<sup>1</sup>[Online]. Available: <https://www.med.upenn.edu/cbica/brats2020/>

<sup>2</sup>[Online]. Available: <http://medicaldecathlon.com/>



where  $h(A, B) = \max_{a \in A} \{\min_{b \in B} \|a - b\|_2\}$ , and  $\|\cdot\|_2$  denotes Euclidean distance.

*Sensitivity (Sens)*, also known as Recall, measures the proportion of actual positives correctly identified by a model. It is calculated as:

$$\text{Sensitivity} = \frac{TP}{TP + FN}. \quad (13)$$

## B. Implementation Details

During training, we applied various data pre-processing and augmentation techniques, including min-max normalization, random crop, random flip, channel shuffling, and additive noise. Given the substantial margins in the original volumes, we first performed coarse cropping to remove outer voxels with a value of zero. Subsequently, random cropping and padding were applied to obtain the processed volumes with fixed sizes of  $128 \times 128 \times 128$  for glioma segmentation and  $320 \times 320 \times 32$  for prostate segmentation. The model was trained using the Ranger [47] optimizer with a learning rate of  $10^{-3}$  and a weight decay of  $10^{-5}$ . The total number of training epochs was set to 300. Due to the limitation of VRAM, the batch size was set to 1. The proposed method was implemented in PyTorch and trained on a single NVIDIA GeForce RTX 3090 GPU with 24 GB memory.

## C. Comparison With State-of-The-Art Methods

**1) Comparison Methods:** We compared our method with both CNN-based and Transformer-based segmentation models, using open-source code for most experiments. The code for TransBTSV2 was implemented by the authors based on TransBTS.<sup>3</sup> We adopted nnU-Net and EquiUnet as baselines. The details of each comparison method are listed below.

- U-Net [4] is the first to employ an encoder-decoder architecture with a fully convolutional network.
- V-Net [8] is a 3D convolutional neural network optimized with Dice loss for volumetric medical image segmentation.
- EquiUnet [44] utilizes self-ensembled, deep-supervised 3D U-Net networks.
- KiUnet [14] combines overcomplete convolutions with U-Net to enhance the accuracy of detecting fine structures and boundaries in images.
- nnU-Net [36] is a self-adapting method for biomedical image segmentation that automatically determines the optimal U-Net architecture and training strategy.
- TransBTS [23] integrates a Transformer with a 3D CNN to improve 3D MRI brain tumor segmentation.
- UNETR [25] employs Transformers in a U-shaped architecture for enhanced 3D medical image segmentation.
- SwinUNETR [26] utilizes SwinTransformer [48] and self-supervised learning in UNETR.
- TransBTSV2 [24] improves TransBTS by incorporating a Hybrid Deformable Model (HDM).

- MISSU [37] is a self-distillation method for a Transformer-based U-Net equipped with a multi-scale fusion block to refine fine-grained details.

**2) BraTS2020:** Since the labels of the BraTS2020 test set is unavailable, the results of testing set need to be evaluated online. For better comparisons, we evaluate our GFNet on BraTS2020 in two ways.

**a) Validation set:** The training set of BraTS2020 is randomly separated into training set and validation set with a ratio of 4 : 1 (296 cases for training and 74 cases for validation). We report the results on BraTS2020 dataset in Table I in the first evaluation way. Due to variations in pre-processing operations that can significantly impact segmentation performance, the compared methods (U-Net, V-Net, EquiUnet, KiUnet) employ identical data pre-processing and augmentation procedures [44], whereas the results of TransBTS, UNETR, SwinUNETR, TransBTSV2, MISSU, and nnU-Net are obtained using their respective official public codes. Moreover, we utilize the same loss functions for training to demonstrate the superiority of our proposed method. We performed a systematic evaluation of GFNet and several state-of-the-art methods. The proposed method named GFNet (E), based on EquiUnet, achieves superior performance, attaining a new average DSC of 84.39% across various glioma areas. For the different areas (i.e., ET, TC, and WT), we also obtain the competitive results in three indicators that outperform the baselines by 2.48%, 5.47 mm and 2.07%, respectively. GFNet (N), based on nnU-Net, achieves the second place with the indicator of Average DSC 83.74%. Further, it should be observed that UNETR, SwinUNETR and nnU-Net do not perform well in segmenting the enhancing tumor region. This may be linked to the data-augmentation by [44], which can address the importance of Enhanced Tumor. Compared to recent Transformer-based methods such as TransBTSV2 and MISSU, our method outperforms them across all target areas. Specifically, GFNet (E) delivers a performance increase of 2.81% over TransBTSV2 and 2.84% over MISSU.

Fig. 4 shows the box plots of the indicators (DSC, HD95 and Sens) for each volume for demonstrating the locality, spread and skewness groups of the segmentation results on the validation set. The first row, including Fig. 4(a)–(c), denotes the box plots of the DSC scores of different glioma areas (ET, TC and WT). The second and third rows, including Fig. 4(d)–(f) and (g)–(i), represent the indicators of HD95 and Sensitivity. For each box plot, the horizontal and vertical coordinates represent the corresponding indicators and different methods, respectively. It is obvious that the boxes of GFNet (N) and GFNet (E) are shorter than those of others, which means the segmentation results of different volumes are more centralized. In other words, the segmentation performance of our method is more stabilized and robust.

**b) Test set:** We train our GFNet on the whole training set and get the predicted segmentation maps of testing images, then upload them to the official website<sup>4</sup> for online evaluation. The evaluation results on the BraTS2020 test set are shown in Table II. Compared to the other methods, GFNet, based

<sup>3</sup>[Online]. Available: <https://github.com/Rubics-Xuan/TransBTS>

<sup>4</sup>[Online]. Available: <https://ipp.cbica.upenn.edu/>

TABLE I

COMPARISON WITH OTHER SEGMENTATION ALGORITHMS ON THE BRATS2020 VALIDATION SET WITH THE INDICATORS OF DICE SIMILARITY COEFFICIENT (%), HD95 (mm) AND SENSITIVITY (%)

| Methods               | Enhanced Tumor (ET) |       |       | Tumor Core (TC) |       |       | Whole Tumor (WT) |       |       | Average      |              |              |
|-----------------------|---------------------|-------|-------|-----------------|-------|-------|------------------|-------|-------|--------------|--------------|--------------|
|                       | DSC↑                | HD95↓ | Sens↑ | DSC↑            | HD95↓ | Sens↑ | DSC↑             | HD95↓ | Sens↑ | DSC↑         | HD95↓        | Sens↑        |
| U-Net ('15) [4]       | 72.02               | 45.86 | 75.20 | 80.13           | 18.08 | 81.82 | 87.89            | 14.86 | 88.18 | 80.01        | 26.27        | 81.73        |
| V-Net ('16) [8]       | 73.56               | 32.53 | 77.04 | 83.03           | 10.31 | 83.16 | 87.91            | 22.81 | 91.41 | 81.50        | 21.88        | 83.89        |
| EquiUnet ('20) [44]   | 77.41               | 29.19 | 79.32 | 80.68           | 15.43 | 82.83 | 87.65            | 23.28 | 91.82 | 81.91        | 22.63        | 84.66        |
| KiUnet ('21) [14]     | 77.25               | 36.15 | 78.16 | 82.55           | 10.19 | 84.41 | 87.75            | 21.64 | 93.76 | 82.52        | 22.66        | 85.44        |
| nnU-Net ('21) [36]    | 75.25               | 36.34 | 77.05 | 83.24           | 11.86 | 81.67 | 90.78            | 12.44 | 93.30 | 83.09        | 20.22        | 84.01        |
| TransBTS ('21) [23]   | 73.86               | 32.07 | 73.26 | 79.85           | 14.16 | 77.21 | 89.66            | 16.32 | 87.76 | 81.12        | 20.85        | 79.41        |
| UNETR ('22) [25]      | 73.01               | 41.10 | 74.49 | 80.63           | 12.97 | 81.76 | 89.76            | 19.27 | 89.86 | 81.13        | 24.45        | 82.04        |
| SwinUNETR ('22) [26]  | 72.84               | 40.68 | 72.60 | 83.20           | 9.31  | 82.30 | 91.11            | 11.73 | 91.28 | 82.38        | 20.57        | 82.06        |
| TransBTSV2 ('22) [24] | 75.17               | 34.61 | 79.00 | 82.27           | 16.56 | 83.63 | 88.79            | 20.41 | 90.57 | 82.08        | 23.86        | 84.40        |
| MISSU ('23) [37]      | 73.62               | 41.03 | 73.11 | 82.15           | 16.99 | 80.33 | 90.42            | 11.92 | 88.65 | 82.06        | 23.31        | 80.69        |
| GFNet (N)             | 74.70               | 34.62 | 75.01 | 84.76           | 10.20 | 84.57 | 91.77            | 8.51  | 91.05 | 83.74        | 17.77        | 83.54        |
| GFNet (E)             | 78.92               | 29.53 | 82.64 | 83.91           | 11.60 | 86.35 | 90.32            | 10.34 | 91.21 | <b>84.39</b> | <b>17.16</b> | <b>86.73</b> |

"Average" denotes the average results of Enhanced Tumor, Tumor Core and Whole Tumor. GFNet (N) and GFNet (E) represent our methods with two different baselines, i.e., nnU-Net and EquiUnet. The best results are highlighted in bold font.

TABLE II

COMPARISON WITH OTHER SEGMENTATION ALGORITHMS ON THE BRATS2020 TEST SET WITH THE INDICATORS OF DICE SIMILARITY COEFFICIENT (%), HD95 (mm) AND SENSITIVITY (%)

| Methods               | Enhanced Tumor (ET) |       |       | Tumor Core (TC) |       |       | Whole Tumor (WT) |       |       | Average      |             |              |
|-----------------------|---------------------|-------|-------|-----------------|-------|-------|------------------|-------|-------|--------------|-------------|--------------|
|                       | DSC↑                | HD95↓ | Sens↑ | DSC↑            | HD95↓ | Sens↑ | DSC↑             | HD95↓ | Sens↑ | DSC↑         | HD95↓       | Sens↑        |
| U-Net ('15) [4]       | 61.26               | 52.00 | 63.77 | 71.72           | 17.63 | 70.26 | 82.38            | 9.39  | 81.30 | 71.79        | 26.34       | 71.78        |
| V-Net ('16) [8]       | 71.48               | 34.90 | 66.85 | 74.55           | 25.67 | 70.25 | 85.24            | 15.05 | 84.37 | 77.09        | 25.21       | 73.82        |
| EquiUnet ('20) [44]   | 74.09               | 31.61 | 76.88 | 79.75           | 11.60 | 80.07 | 87.88            | 8.00  | 94.12 | 80.58        | 17.07       | 83.69        |
| KiUnet ('21) [14]     | 74.96               | 31.11 | 75.81 | 81.18           | 9.35  | 83.32 | 88.57            | 7.38  | 92.17 | 81.57        | 15.95       | 83.77        |
| nnU-Net ('21) [36]    | 72.38               | 39.15 | 73.23 | 83.59           | 6.17  | 80.20 | 90.19            | 5.04  | 90.88 | 82.04        | 16.79       | 81.44        |
| TransBTS ('21) [23]   | 78.73               | 17.95 | -     | 81.73           | 9.77  | -     | 90.09            | 4.96  | -     | 83.52        | 10.89       | -            |
| UNETR ('22) [25]      | 70.36               | 41.39 | 70.70 | 79.74           | 11.68 | 81.16 | 89.42            | 6.93  | 90.72 | 79.84        | 20.00       | 80.86        |
| SwinUNETR ('22) [26]  | 71.21               | 42.96 | 72.06 | 81.15           | 10.10 | 79.79 | 89.72            | 8.28  | 91.10 | 80.69        | 20.45       | 80.98        |
| TransBTSV2 ('22) [24] | 78.53               | 18.41 | 76.73 | 82.44           | 8.70  | 81.53 | 90.04            | 5.26  | 91.41 | 83.67        | 10.79       | 83.22        |
| MISSU ('23) [37]      | 74.58               | 27.70 | 74.70 | 81.52           | 11.84 | 79.63 | 89.71            | 5.52  | 89.64 | 81.94        | 15.02       | 81.32        |
| GFNet (N)             | 73.36               | 36.23 | 73.53 | 84.81           | 5.87  | 82.63 | 90.59            | 4.63  | 91.00 | 83.01        | 15.58       | 82.39        |
| GFNet (E)             | 79.90               | 17.50 | 78.97 | 83.72           | 6.48  | 81.85 | 90.10            | 6.00  | 90.21 | <b>84.57</b> | <b>9.99</b> | <b>83.67</b> |

"Average" denotes the average results of Enhanced Tumor, Tumor Core and Whole Tumor. GFNet (N) and GFNet (E) represent our methods with two different baselines, i.e., nnU-Net and EquiUnet. The best average results are highlighted in bold font.

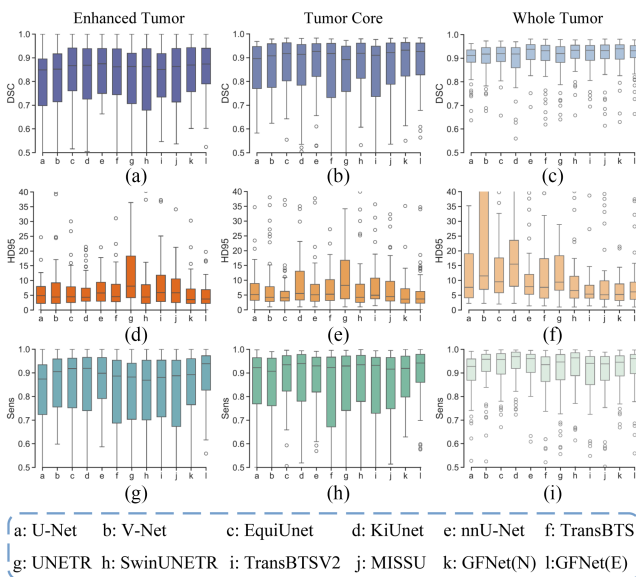


Fig. 4. Box plots of experimental results on the validation set of BraTS2020. Different rows represent DSC, HD95, and Sens, respectively. Different columns denote various segmentation areas, i.e., Enhanced Tumor, Tumor Core and Whole Tumor.

on EquiUnet, yields the best results in terms of average DSC and HD95. Meanwhile, GFNet (N) can effectively improve the performance of nnU-Net. However, KiUnet obtains the best result of average Sensitivity, indicating that the method is capable of correctly identifying the majority of the glioma areas. Compared to KiUnet, our GFNet (E) exhibits only a tiny discrepancy with the value of 0.1%. The indicator of Sensitivity is absented in [23]. GFNet (N) outperforms TransBTSV2 and MISSU with improvements of 1.08% and 3.21% in average DSC, respectively.

3) *Glioma Segmentation Task of MSD*: We also evaluate our method and other comparison methods on the glioma segmentation task of MSD dataset. The quantitative results are reported in Table III. Our GFNet (N) can achieve the best average segmentation results on various regions of gliomas with the DSC, HD95 and Sens of 83.09%, 9.72 mm and 81.59%. GFNet (E) and GFNet (N) can surpass their baselines in varying degrees. This indicates our approach pays more attention to the edge of glioma with the help of dual-path gradient-guided training strategy and Fourier Edge-enhancement Module. It also should be noticed that GFNet (N) do not surpass the baseline nnU-Net for the segmentation of Tumor Core and Whole Tumor with the indicator of DSC, but the differences are fairly small with



TABLE III

COMPARISON WITH OTHER SEGMENTATION ALGORITHM ON GLIOMA SEGMENTATION DATASET OF MSD WITH THE INDICATORS OF DICE SIMILARITY COEFFICIENT (%), HD95 (mm) AND SENSITIVITY (%)

| Methods               | Enhanced Tumor (ET) |       |       | Tumor Core (TC) |       |       | Whole Tumor (WT) |       |       | Average      |             |              |
|-----------------------|---------------------|-------|-------|-----------------|-------|-------|------------------|-------|-------|--------------|-------------|--------------|
|                       | DSC↑                | HD95↓ | Sens↑ | DSC↑            | HD95↓ | Sens↑ | DSC↑             | HD95↓ | Sens↑ | DSC↑         | HD95↓       | Sens↑        |
| U-Net ('15) [4]       | 70.38               | 23.03 | 74.87 | 77.06           | 24.25 | 79.96 | 87.64            | 16.16 | 89.67 | 78.36        | 21.15       | 81.50        |
| V-Net ('16) [8]       | 74.76               | 11.94 | 79.83 | 79.71           | 17.49 | 81.42 | 85.05            | 29.62 | 94.10 | 79.84        | 19.68       | 85.11        |
| U-Net++ ('19) [49]    | 60.56               | 36.83 | 60.80 | 65.50           | 29.35 | 61.60 | 77.34            | 26.70 | 77.19 | 67.80        | 30.96       | 66.53        |
| EquiUnet ('20) [44]   | 75.04               | 11.38 | 83.03 | 80.60           | 16.22 | 84.27 | 87.41            | 24.60 | 95.01 | 81.02        | 17.40       | 87.44        |
| KiUnet ('21) [14]     | 74.28               | 18.78 | 87.65 | 79.13           | 15.48 | 84.61 | 87.38            | 28.96 | 94.02 | 80.26        | 21.08       | <b>88.76</b> |
| nnU-Net ('21) [36]    | 74.77               | 18.89 | 75.06 | 82.69           | 7.40  | 80.00 | 90.37            | 9.12  | 89.61 | 82.61        | 11.80       | 81.56        |
| UNETR ('22) [25]      | 74.34               | 18.16 | 78.81 | 78.66           | 13.09 | 80.19 | 89.04            | 15.58 | 91.04 | 80.68        | 15.61       | 83.35        |
| SwinUNETR ('22) [26]  | 75.87               | 16.47 | 78.62 | 81.62           | 9.86  | 81.11 | 90.08            | 10.25 | 90.76 | 82.52        | 12.19       | 83.50        |
| TransBTSV2 ('22) [24] | 74.21               | 24.91 | 78.99 | 79.20           | 22.52 | 79.21 | 86.35            | 16.95 | 87.24 | 79.92        | 21.46       | 81.81        |
| MISSU ('23) [37]      | 75.36               | 20.61 | 77.32 | 79.64           | 19.94 | 76.57 | 89.27            | 13.63 | 88.41 | 81.42        | 18.06       | 80.77        |
| GFNet (N)             | 76.32               | 14.70 | 76.53 | 82.66           | 6.32  | 79.53 | 90.29            | 8.13  | 88.71 | <b>83.09</b> | <b>9.72</b> | 81.59        |
| GFNet (E)             | 76.46               | 9.72  | 84.89 | 81.63           | 13.56 | 86.56 | 88.05            | 18.12 | 93.16 | 82.05        | 13.80       | 88.21        |

"Average" denotes the average results of Enhanced Tumor, Tumor Core and Whole Tumor. GFNet (N) and GFNet (E) represent our methods with two different baselines, i.e., nnU-Net and EquiUnet. The best average results are highlighted in bold font.

TABLE IV

COMPARISON WITH OTHER SEGMENTATION ALGORITHMS ON PROSTATE SEGMENTATION DATASET OF MSD WITH THE INDICATORS OF DICE SIMILARITY COEFFICIENT (%), HD95 (mm) AND SENSITIVITY (%)

| Methods               | Peripheral Zone (PZ) |       |       | Transition Zone (TZ) |       |       | Average      |             |              |
|-----------------------|----------------------|-------|-------|----------------------|-------|-------|--------------|-------------|--------------|
|                       | DSC↑                 | HD95↓ | Sens↑ | DSC↑                 | HD95↓ | Sens↑ | DSC↑         | HD95↓       | Sens↑        |
| U-Net ('15) [4]       | 58.41                | 6.32  | 69.60 | 80.95                | 5.84  | 84.84 | 69.68        | 6.08        | 77.22        |
| V-Net ('16) [8]       | 60.34                | 5.83  | 73.35 | 84.55                | 3.05  | 84.60 | 72.45        | 4.44        | 78.97        |
| U-Net++ ('19) [49]    | 60.79                | 5.78  | 65.37 | 83.46                | 3.37  | 84.00 | 72.12        | 4.58        | 74.68        |
| EquiUnet ('20) [44]   | 63.03                | 6.34  | 70.06 | 85.21                | 2.95  | 84.05 | 74.12        | 4.65        | 77.06        |
| KiUnet ('21) [14]     | 61.55                | 6.55  | 64.00 | 84.25                | 4.15  | 88.28 | 72.90        | 5.35        | 76.14        |
| nnU-Net ('21) [36]    | 72.06                | 4.49  | 80.17 | 88.90                | 2.46  | 86.40 | 80.48        | 3.48        | 83.29        |
| UNETR ('22) [25]      | 63.80                | 6.81  | 67.91 | 86.31                | 3.40  | 83.85 | 75.06        | 5.11        | 75.88        |
| SwinUNETR ('22) [26]  | 64.06                | 6.74  | 66.26 | 87.13                | 3.58  | 89.13 | 75.59        | 5.16        | 77.70        |
| TransBTSV2 ('22) [24] | 61.24                | 5.41  | 60.18 | 87.73                | 1.31  | 90.29 | 74.48        | 3.36        | 75.23        |
| MISSU ('23) [37]      | 61.27                | 5.78  | 70.04 | 85.10                | 2.05  | 84.29 | 73.19        | 3.72        | 77.17        |
| GFNet (N)             | 72.58                | 4.61  | 81.45 | 89.41                | 2.54  | 87.21 | <b>81.00</b> | <b>3.58</b> | <b>84.33</b> |
| GFNet (E)             | 65.49                | 5.92  | 74.14 | 86.74                | 2.58  | 84.61 | 76.12        | 4.25        | 79.38        |

"Average" denotes the average results of Enhanced Tumor, Tumor Core and Whole Tumor. GFNet (N) and GFNet (E) represent our methods with two different baselines, i.e., nnU-Net and EquiUnet. The best average results are highlighted in bold font.

the values of 0.03% and 0.08%. Compared to recent works like TransBTSV2 and MISSU, GFNet (N) brings 3.97% and 2.05% improvements in average DSC, respectively.

4) *Prostate Segmentation Task of MSD*: We further evaluate the performance of prostate segmentation on the MSD dataset, with quantitative results summarized in Table IV. Our GFNet consistently outperforms the backbones (EquiUnet and nnU-Net) and other competing methods. Specifically, our method achieves an average DSC improvement of 9.24% over vanilla U-Net [4]. Compared to other CNN-based methods, such as V-Net [8], U-Net++ [49], and KiUnet [14], our method surpasses them with a large margin, due to the edge and global information extracted by DGT and FEM. While Transformer based methods, including UNETR [25], SwinUNETR [26], TransBTSV2 [24] and MISSU [37], capture long-range dependencies for global information extraction, they struggle with prostate boundary and local-related information extraction. GFNet achieves superior performance compared to Transformer-based methods. Finally, integrating DGT and FEM into the baselines, nnU-Net [36] and EquiUnet [44], yields an average DSC improvement of 0.6% and 0.27%, respectively. Furthermore, GFNet enhances segmentation performance in the Peripheral Zone and Transition Zone. The results in Table IV demonstrate that the proposed GFNet

not only excels in glioma segmentation but also significantly improves the prostate segmentation performance.

#### D. Qualitative Analysis

We compare the visualization results of glioma segmentation utilizing various methods, including KiUnet [14], EquiUnet [44], nnU-Net [36], TransBTS [23], SwinUNETR [26], TransBTSV2 [24], MISSU [37], and our methods based on nnU-Net and EquiUnet in Fig. 5. The first and last columns represent the input images and labels, respectively. The number at the top left of each image indicates the current slice and we also show the average DSC of three different segmentation areas at the bottom of each image. The enlarges local areas are shown at the top right of slices. Benefited from Dual-path Gradient-guided Training strategy (DGT) and Fourier Edge-enhancement Module (FEM) which can preserve detailed information, it is evident in that our GFNet can segment the objective regions more accurately.

#### E. Ablation Study

1) *Efficiency of Each Module*: The proposed method is composed of two technical parts, i.e., Dual-path Gradient-guided

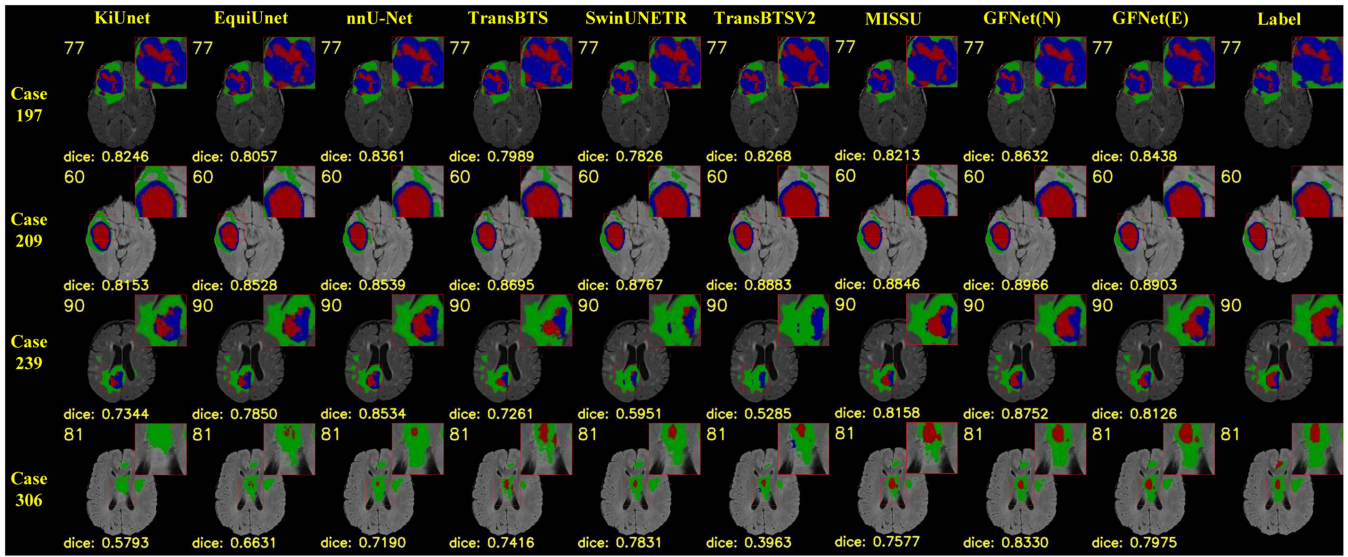


Fig. 5. Qualitative results of BraTS2020 validation dataset. Average DSC are illustrated at the bottom of the images. The number of slices is shown on the top left of the images. Three classes of necrotic and non-enhancing tumor (Label 1), peritumoral edema (Label 2) and GD-enhanced tumor (Label 4) are highlighted with different colors.

TABLE V

ABLATION EXPERIMENTS ON THE BRA TS2020 VALIDATION SET WITH THE INDICATOR OF DSC (%)

| Methods   | FEM | DGT | ET    | TC    | WT    | Avg.  |
|-----------|-----|-----|-------|-------|-------|-------|
| GFNet (E) | ✓   |     | 77.41 | 80.68 | 87.45 | 81.91 |
|           |     |     | 78.63 | 82.17 | 90.20 | 83.66 |
|           | ✓   | ✓   | 78.64 | 83.27 | 89.16 | 83.69 |
|           |     | ✓   | 78.92 | 83.91 | 90.32 | 84.39 |
| GFNet (N) | ✓   |     | 75.25 | 83.24 | 90.78 | 83.08 |
|           |     |     | 75.69 | 83.51 | 90.41 | 83.29 |
|           | ✓   | ✓   | 76.01 | 83.42 | 90.37 | 83.27 |
|           |     | ✓   | 74.70 | 84.76 | 91.77 | 83.74 |

Training strategy (DGT) and Fourier Edge-enhancement Module (FEM). In Table V, we perform our GFNet on two popular baselines EquiUnet and nnU-Net. The results in the second and sixth rows represent the performance of two baseli en method (EquiUnet and nnU-Net). Subsequently, we incorporate the FEM and DGT modules into the two methods. Both methods achieved varying degrees of performance improvement. The performance improvement of EquiUnet is quite significant, while nnU-Net experiences an enhancement of only 0.66%. We hypothesize that the modest enhancement in performance is due to nnU-Net utilizing a data preprocessing approach tailored to its network structure; altering the network structure may render the preprocessed data less suitable, thereby resulting in a less improvement.

2) *The Necessity of FEM*: We also conducted an experiment where GFNet (E) was used without FEM during inference on the BraTS2020 validation set, as shown in Table VI. The results indicate that “GFNet (E) inference w/o FEM” results in performance degradation in TC, WT, and Avg, demonstrating the importance of FEM during inference.

TABLE VI

DSC (%) RESULTS FOR GFNET (E) INFERENCE WITH AND WITHOUT FEM ON THE BRA TS2020 VALIDATION SET

| Method                      | ET    | TC    | WT    | Avg.  |
|-----------------------------|-------|-------|-------|-------|
| GFNet (E) inference w/o FEM | 79.03 | 82.31 | 88.84 | 83.39 |
| GFNet (E) inference w FEM   | 78.92 | 83.91 | 90.32 | 84.39 |

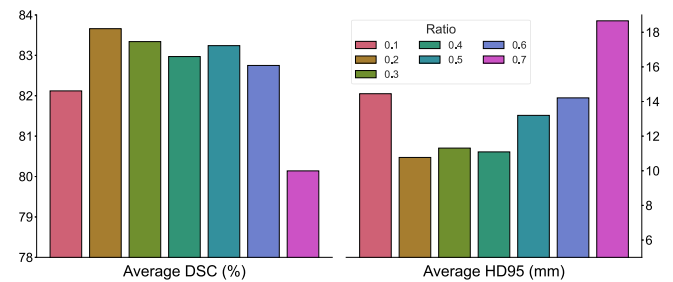


Fig. 6. Influence of window size. The left and right sub-figures represent Average DSC (%) and HD95 (mm), respectively. Different colors denote the various values of  $r$ .

3) *Influence of Window Size*: In Fourier Edge-enhancement Module, a cropping window is used to separate the high and low-frequency components of the amplitude spectrum. The choice of window size is crucial, having a significant impact on the extraction of edge information. The ratio  $r$  is defined as the quotient of the window size  $S_{win}$  by the feature size  $S_f$ , expressed as  $r = S_{win}/S_f$ . Extensive experiments with various ratios  $r$  were conducted on the BraTS2020 validation set, specifically for  $r$  values ranging from 0.1 to 0.7. Note that when  $r = 0$ , the FEM extracts all frequency components. However, as the value of  $r$  increases to 0.7, the DSC and HD95 indicators drop significantly. It was determined that an optimal  $r$  value is 0.2. Detailed results are shown in Fig. 6, where the left and

TABLE VII

COMPARISON OF THE NUMBER OF TRAINABLE PARAMETERS AND GFLOPS

| Methods                 | Trainable Params (M) | GFLOPs  |
|-------------------------|----------------------|---------|
| U-Net ('15) [4]         | 9.43                 | 45.28   |
| V-Net ('16) [8]         | 45.72                | 1023.21 |
| U-Net++ ('19) [49]      | 109.85               | 8059.34 |
| EquiUnet ('20) [44]     | 23.15                | 997.77  |
| KiUnet ('21) [14]       | 9.47                 | 292.17  |
| nnU-Net ('21) [36]      | 27.66                | 450.60  |
| TransBTS ('21) [23]     | 30.62                | 263.73  |
| UNETR ('22) [25]        | 102.06               | 203.32  |
| SwinUNETR ('22) [26]    | 61.99                | 793.92  |
| TransBTSV2 ('22) [24]   | 30.77                | 289.75  |
| MISSU (23) [37]         | 30.62                | 263.73  |
| GFNet (N)               | 27.66                | 453.26  |
| GFNet (E)               | 23.15                | 1000.43 |
| GFNet (N) <sup>Tr</sup> | 27.66                | 906.52  |
| GFNet (E) <sup>Tr</sup> | 23.15                | 2000.86 |

right vertical axes represent DSC and HD95, respectively, and different colors indicate distinct  $r$  values.

## V. DISCUSSION

### A. Model Trainable Parameters and GFLOPs

We also calculate the number of parameters and Giga Floating Point Operations Per Second (GFLOPs) for various models, as presented in Table VII. EquiUnet [44] and nnU-Net [36] serve as the baselines for our GFNet. For a fair comparison, The GFLOPs's value of each method is computed with the same input volume with the size of  $4 \times 128 \times 128 \times 128$ . Our method scarcely introduces additional training parameters or computational complexity during inference phase. Because we utilized a Siamese network architecture to address important regions using gradient information, the GFLOPs of GFNet have doubled for a single iteration during the training phase. GFNet (N)<sup>Tr</sup> and GFNet (E)<sup>Tr</sup> represent the number of trainable parameters (M) and GFLOPs for one iteration of GFNet based on nnU-Net and EquiUnet during the training phase. GFNet (N) and GFNet (E) denote the number of trainable parameters (M) and GFLOPs for one iteration of GFNet based on nnU-Net and EquiUnet during the inference phase.

### B. Visualization of Feature and Gradient Maps

In the experiment section, we have shown the quantitative results, demonstrating the effectiveness of "GFNet", they cannot explain the efficacy of our method from the aspect of feature representations. Thus we visualize the middle-layer feature maps, corresponding gradients, and Fourier edge-enhancement feature maps in Fig. 7. The first row shows the original feature maps from the middle layers of Encoder, with their values scaled to the range of 0 to 255. 1) To assess the benefits of gradients for network training, we visualize both the gradients and gradient-enhance feature maps. The visualized gradients  $\frac{\partial L}{\partial f_i^e}$  ( $i \in [1, 2, 3, 4]$ ) are demonstrated in the second row of Fig. 7. It is obvious that the gradient maps capture the information of the segmented targets (highlighted area). Additionally, the edges in the gradient maps are more defined than in the feature maps, and the contrast between the foreground and background is more

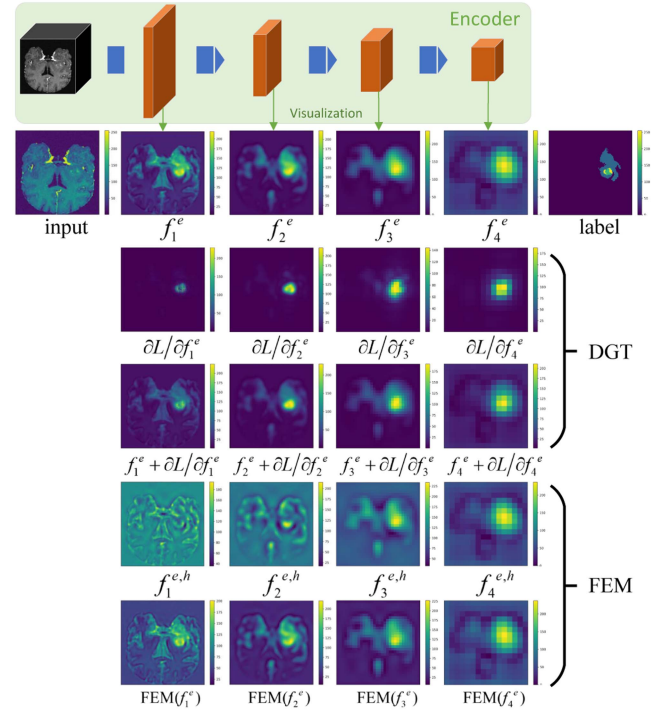


Fig. 7. Visualization of feature and gradient maps. Row 1 represents the visualized feature maps. Row 2 and 3 denote the visualized gradient maps and the gradient-enhanced feature maps in DGT. Row 4 and 5 show the high-frequency part and edge-enhanced feature maps in FEM.

TABLE VIII

DSC (%) RESULTS FOR DIFFERENT HIGH-PASS FILTERS ON THE BRATS2020 VALIDATION SET

| Filters     | ET    | TC    | WT    | Avg.  |
|-------------|-------|-------|-------|-------|
| Butterworth | 79.58 | 84.56 | 89.02 | 84.39 |
| Gaussian    | 77.35 | 84.59 | 88.86 | 83.60 |
| Exponential | 79.14 | 84.40 | 89.82 | 84.46 |
| Bessel      | 77.93 | 84.16 | 89.59 | 83.89 |
| Random      | 78.66 | 82.84 | 88.40 | 83.30 |
| Binary      | 78.92 | 83.91 | 90.32 | 84.39 |

pronounced. We then combine the gradient information with the feature maps. The third row shows the sum of feature maps and their corresponding gradients  $f_i^e + \frac{\partial L}{\partial f_i^e}$  ( $i \in [1, 2, 3, 4]$ ). Notably, leveraging the gradients can enhance the feature maps for better foreground and background discrimination. 2) For the proposed FEM, the fourth row represents the high-frequency part of the feature map  $f_i^{e,h}$  ( $i \in [1, 2, 3, 4]$ ). While the last row represents the output feature maps of FEM ( $f_i^e$ ). Compared to the feature maps in Row one, the edges of the Fourier edge-enhanced features in the last row are more distinct. The visualization results can demonstrate the effectiveness of FEM and DGT.

### C. Impact of Different High-Pass Filters in FEM

In FEM, we utilize high-pass filters to remove low-frequency information, thus isolating high-frequency edge areas. While we primarily use binary high-pass filters in FEM, we also tested smoother filters—Butterworth, Gaussian, Exponential, and Bessel—to evaluate the impact of filters. Validation on the BraTS2020 dataset, as shown in Table VIII, reveals that FEM



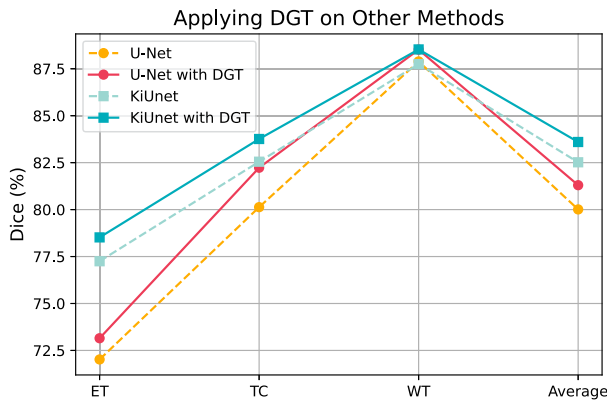


Fig. 8. The quantitative results of U-Net and KiUnet, and their improved versions with DGT.

with an Exponential high-pass filter achieves the best performance, with an average DSC of 84.46%, surpassing the binary filter. The Butterworth high-pass filter neither improves nor worsens performance, while the Gaussian and Bessel filters cause a slight decline. Overall, the choice of different filters does affect the results, but the impact is quite limited. Furthermore, replacing the binary high-pass filter with a parameter-randomized filter did not yield the best segmentation performance.

#### D. Applying DGT on Other Methods

To further validate the effectiveness of DGT, we applied it to other models. Specifically, we used U-Net [4] and KiUnet [14] as baselines and incorporated the proposed DGT for comparison. The results on the validation set of BraTS2020 dataset are shown in Fig. 8. The horizontal axis represents different segmentation regions, namely ET, TC, WT, and their average, while the vertical axis indicates the DSC score. U-Net and KiUnet are represented by light red and light blue, respectively. Solid lines denote the methods with DGT, and dashed lines represent their corresponding baselines. As shown in Fig. 8, it is obvious that DGT leads to performance improvements, particularly for ET and TC segmentation. However, it does not significantly enhance WT segmentation, likely due to the clearer and easier-to-segment boundaries of the whole tumor region, which limit the potential gains from DGT. Specifically, U-Net with DGT improved DSC by 1% and 2.6% for ET and TC, respectively, while KiUnet with DGT showed improvements of 1.6% and 1.5%. These experimental results demonstrate that DGT effectively enhances the performance of different models.

#### E. Limitations and Future Work

While the proposed GFNet significantly improves segmentation performance in a straightforward and efficient manner with minimal additional computational cost during inference, this study has certain limitations. The use of a Siamese network for training doubles the GFLOPs compared to the baseline, making it challenging to train the model on devices with limited computational resources. Given the exponential growth in data and computational capabilities, there has been a surge in universal

models [50], [51], [52] capable of segmenting various targets using image or text prompts. However, GFNet is specifically tailored for certain targets, which limits its versatility. Additionally, current universal models often fail to meet clinical standards due to their unstable performance, a critical drawback in medical applications.

In future research, we plan to reduce computational costs during training, with a focus on how to obtain gradient information. Currently, in DGT, gradients are obtained from another branch of the Siamese network. An alternative approach, such as using gradients from previous epochs, may also enhance glioma segmentation performance. This represents a promising direction for GFNet. Additionally, we aim to integrate our proposed method with universal models to enhance its generalization and versatility. By combining the strengths of GFNet with the flexibility of universal models, we hope to develop a more robust solution for a wider range of segmentation tasks. Finally, we will validate our GFNet in various scenarios, including diverse medical imaging datasets, to thoroughly evaluate its performance and clinical applicability.

## VI. CONCLUSION

In this paper, we propose a dual-path gradient-guided training strategy with Fourier edge-enhancement for glioma segmentation in multimodal 3D MRI. The novel training strategy under the Dual-path Gradient-guided (DGT) takes full advantage of intrinsic gradient information to focus on the areas with large errors. We further put forward a Fourier Edge-enhancement Module (FEM) to effectively capture global contextual information of the feature maps. To validate the effectiveness of our method, we conduct the confirmatory experiments on BraTS2020 and glioma and prostate segmentation of MSD datasets. In conclusion, our solution to glioma segmentation shows promising results and has potential to promote clinical workflow with slight computational cost increasing.

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